12. ARC Safety Peak Limiter & Dynamic Gain Smoothing Math

Time Dilation DAW Technical Proof Series (v0.0.1)

1. Overview & Ceiling Threshold Math

In relativistic time-dilated audio synthesis, dynamic time warping and feedback cascades can produce amplitude spikes.

To prevent digital clipping, our – DuoNode) incorporates an ARC Safety Peak Limiter operating at -1.5 dBFS ceiling.

$$C = 10^{\frac{-1.5}{20}} \approx 0.841395$$

For stereo audio blocks, the instantaneous peak magnitude $p[n]$ and target gain $g_{target}[n]$ are computed as:

2. Attack & Adaptive ARC Release Dynamics

$$g_{target}[n] = \min\left(1.0, \frac{C}{\max(p[n], \epsilon)}\right)$$

Attack time constant = 0.05 ms (50 us) for ultra-fast transient suppression.

Adaptive ARC Release timing dynamically switches based on peak intensity ratio $R = p[n] / C$:

$$\tau_{rel}[R] = \begin{cases} 10 \text{ ms} & \text{for transient spikes (R < 1.5)} \\ 300 \text{ ms} & \text{for sustained peaks (R ≥ 1.5)} \end{cases}$$

Final output audio signals are scaled by $g[n]$, guaranteeing zero hard-clipping at peak ceiling.